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But one minute is 60 second, therefore, expressing time in terms of second

$$f = \frac{15}{60} = \frac{1}{4} = 0.25\,\text{Hz}$$

The time taken by the toy duck for going one time up and down is time period which is inverse of frequency

$$T = \frac{1}{f} = \frac{1}{0.25} = 4\ \text{s}$$

The velocity of ocean wave is

$v = \lambda f = 1.2 \times 0.25 = 0.3\ \text{m s}^{-1}$.

**Amplitude of a wave:**

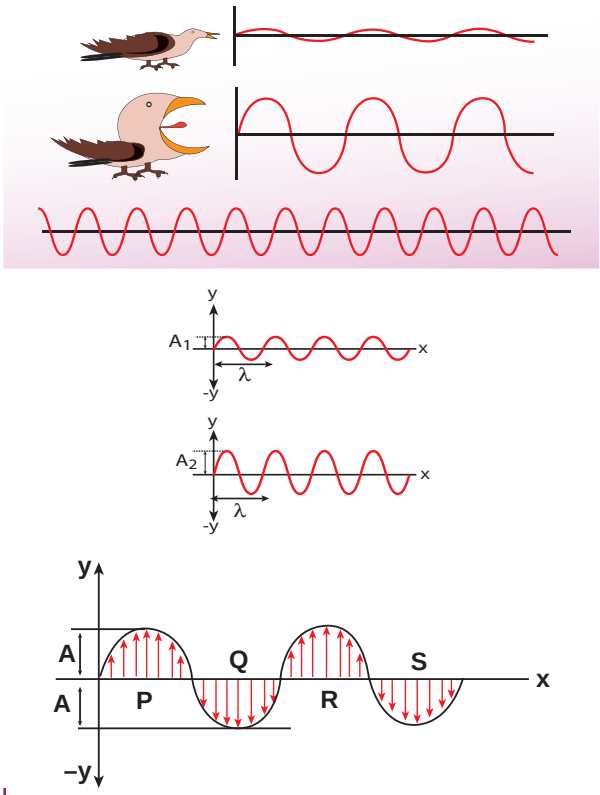
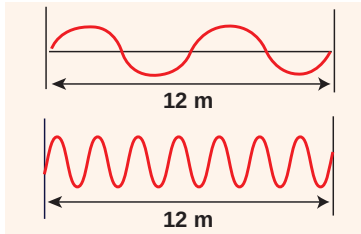



**Figure 11.14** Waves of different amplitude

The waves shown in the Figure 11.14 have same wavelength, same frequency and same time period and also move with same velocity. The only difference between two waves is the height of either crest or trough. This means, the height of the crest or trough also signifies a wave character. So we define a quantity called an amplitude of the wave, as the maximum displacement of the medium with respect to a reference axis (for example in this case x-axis). Here, it is denoted by A.

## EXAMPLE 11.5

Consider a string whose one end is attached to a wall. Then compute the following in both situations given in figure (assume waves crosses the distance in one second)




(a) Wavelength, (b) Frequency and (c) Velocity

**Solution**

| | First case | Second case |
|---|---|---|
| **(a)** Wavelength | $\lambda = 6$ m | $\lambda = 2$ m |
| **(b)** Frequency | $f = 2$ Hz | $f = 6$ Hz |
| **(c)** Velocity | $v = 6 \times 2 = 12\ \text{m s}^{-1}$ | $v = 2 \times 6 = 12\ \text{m s}^{-1}$ |

This means that the speed of the wave along a string is a constant. Higher the frequency, shorter the wavelength and vice versa, and their product is velocity which remains the same.

## 11.3 VELOCITY OF WAVES IN DIFFERENT MEDIA

Suppose a hammer is stroked on long rails at a distance and when a person keeps his ear near the rails at the other end he/she will hear two sounds, at different instants. The sound that is heard through the rails (solid medium)